

Classic Improvements in Power Supplies



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Requirement for continuous improvement in energy efficiency keeps design engineers on the edge as they work hard to come up with better solutions. Test instruments are the key to aid engineers with design-validation and performance testing. Today's power supplies include those designed for a wide variety of uses as well as high-performance options for a variety of applications. These are used in test environments including research and development and production in embedded, semiconductor, nanotechnology and superconductor industries.

"High sourcing accuracy and built-in control functions make these ideal for Hall effect measurements, resistance measurements using delta mode, pulsed measurements and differential-conductance measurements. Programmable pulse widths limit power dissipation in delicate components and support pulsed I-V measurements," says Naresh Narasimhan, country manager, Tektronix India.

So what are the latest improvements seen in these equipment?

FGPA-aided power supplies

A source measure unit (SMU) is a superset of a power supply module that can not only supply voltage or current but also measure and control both current and voltage. It uses closed-loop feedback control to ensure that the desired output voltage is correctly applied to the load under test. Typically, a DC power supply can only source voltage/current. An SMU, on the other hand, can source as well as sink voltage/current so that it can act as a DC supply as well as a controlled load.

In a lot of SMU or power supply applications, you need to give voltage to sensitive circuits like a micro-electro-mechanical system (MEMS). Even small voltages beyond the operating maximum voltages could potentially damage some MEMS sensors. The other side is that it is very difficult to maintain this very stringent voltage level when you are powering up a circuit.

Raviteja Chivukula, technical market-

ing engineer, National Instruments, explains, "Because of resistance, inductance and capacitance present in any circuit, components and wires, any normal DC power supply connected even for a small time could create a small spike. This will be difficult to remove for a normal SMU while ensuring that the rise and test times are small. In our case, online gains of the control unit on the SMU are tuneable by the customer. What happens here is that the SMU supplies voltage and then measures and controls the voltage and current experienced by the device under test (DUT). It will now use its control loop (with custom control loop gains set by the user) to make sure that the DUT does not experience overshoots. Usually this control loop is analogue so is not customisable, but newer models are customisable because the control loop runs on field-programmable gate arrays (FPGAs)."

National Instruments launched PXIe-4139 last year, featuring their SourceAdapt technology to address this issue by helping to custom-tune the SMU response to a given load. This provides an SMU response with minimal settling times (faster test) and eliminates overshoots and oscillations, protecting sensitive DUTs and also helps ensure system stability.

How does this work? Instead of the traditional analogue control loop, these are powered by digital control-loop technology. Because the control loop resides in the digital domain, this feature gives you the programmatic control of critical control-loop parameters, which, in turn, gives you the ability to train the SMU to behave in a certain way, to a specific load. You would typically find ideal control-loop settings for given types of loads or DUTs during the system development process. Once the ideal settings are identified and stored in the control programme, all you need is to know which DUT to test.

"By applying the appropriate settings for that particular DUT, you get the SMU to provide a perfect response without overshoots or oscillations (which is the primary cause of DUT damage for sensitive components and some MEMS sensors) and at the same time, without slowing down the SMU response (hence, faster test time)," adds Chivukula.

Keithley's Model 2460 Source-Meter-brand-

Keithley's Model 2460 SourceMeter

